Tutorial 1

Discrete signals and systems

SC3102: Signals, Systems, and Transforms

Question 1

Periodicity

Periodicity and Sinusoids

- A discrete-time signal is **periodic** if some positive integer N satisfies

$$x[n+N] = x[n] \qquad \text{for every } n \in \mathbb{Z}.$$

- The smallest such N is the **fundamental period**.
- Cosines in particular have the period:

$$\cos(x + 2\pi) = \cos(x)$$

1(a): Which signals are periodic?

- Find the fundamental periods of these real discrete-time signals:

$$x_1[n] = 4\sin\left(\frac{\pi n}{12}\right), \quad x_2[n] = \cos(3.5\pi n),$$

$$x_3[n] = \cos n, \qquad x_4[n] = \sin(\pi n) - \cos n.$$

1(a): The first two signals

- $x_1[n] = 4\sin(\pi n/12)$:

$$\frac{\pi N}{12} = 2\pi m \quad \Longrightarrow \quad \boxed{N = 24}.$$

- $x_2[n] = \cos(3.5\pi n)$:

$$\frac{7\pi N}{2} = 2\pi m \quad \Longrightarrow \quad 7N = 4m \quad \Longrightarrow \quad \boxed{N = 4}.$$

Choose the smallest positive integer N such that m is an integer

1(a): The Signal x_3

- Recall the third signal:

$$x_3[n] = \cos n$$

- A period would require

$$\cos N = \cos 0 = 1 \quad \Longrightarrow \quad N = 2\pi m, \qquad m \in \mathbb{Z}.$$

- No positive integer N satisfies this. The signal is **aperiodic**.

1(a): The Signal x_4

- Finally, recall the fourth signal x_4:

$$x_4[n] = \sin(\pi n) - \cos n$$

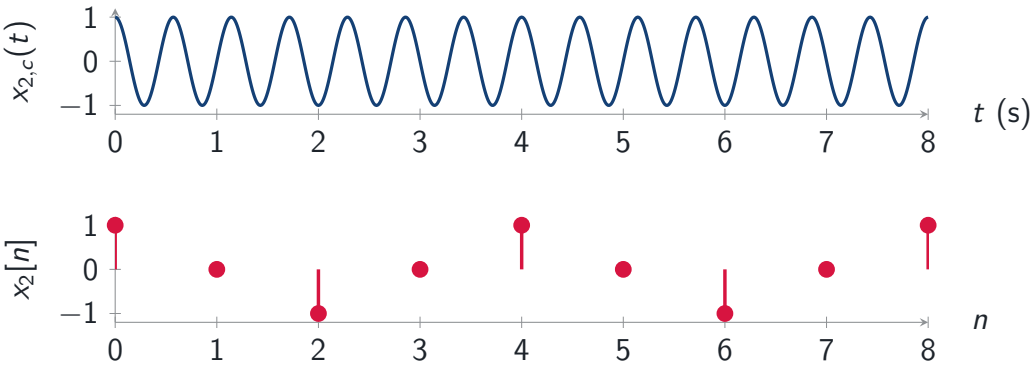
- At integer indices, $\sin(\pi n) = 0$.

$$x_4[n] = -\cos n.$$

- Therefore x_4 is also **aperiodic**.

1(b): Comparing Discrete and Continuous Frequencies

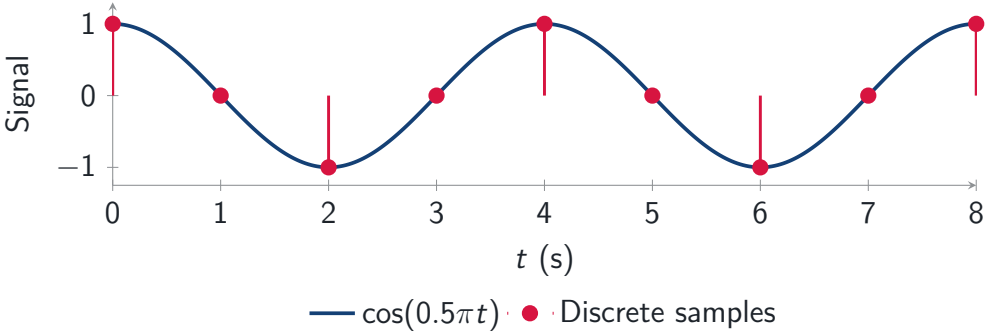
- Lets plot the two signals ($\cos(3.5\pi t)$ and $\cos(3.5\pi n)$)



1(b): Aliasing

- A 0.25 Hz continuous-time sinusoid yields the same samples:

$$\cos(3.5\pi n) = \cos(-0.5\pi n) = \cos(0.5\pi n).$$



1(c): Periodicity of $x_1 + x_2$

- The two signals repeat after 24 and 4 samples.
- Their sum repeats after

$$\boxed{\text{lcm}(24, 4) = 24 \text{ samples.}}$$

1(c): Periodicity of $x_3 + x_4$

- The terms cancel:

$$x_3[n] + x_4[n] = \cos n + \sin(\pi n) - \cos n = 0.$$

- The zero signal is periodic with period 1

1(d): Absolute Summability of Periodic Signals

- Recall the definition of absolute summability:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty.$$

- Can a nonzero periodic signal satisfy this?

1(d): Periodic Signals are not Absolutely Summable

- Suppose x is a periodic signal with period N with $x[k] = a \neq 0$
- Then, we have:

$$\sum_{n=-\infty}^{\infty} |x[n]| \geq \sum_{m=-\infty}^{\infty} |x[k+mN]| = \sum_{m=-\infty}^{\infty} |x[k]| = \infty$$

Non-zero periodic signals are not absolutely summable.

Question 2

Sampling and Aliasing

Sampling

- Let $x_c(t)$ be a continuous-time signal
- We choose a sampling frequency f_sHz ($T_s = 1/f_s$)
- We then sample by evaluating x_c at multiples of T_s:

$$x[n] = x_c(nT_s).$$

Aliasing

- Aliasing is when multiple continuous signals yield the same discrete samples
- For sinusoids in particular recall that:

$$\cos(\theta + 2\pi n) = \cos\theta, \qquad n \in \mathbb{Z}.$$

- Consequently for any Ω and $m \in \mathbb{Z}$:

$$\cos(2\pi mn \pm \Omega n) = \cos((2\pi m \pm \Omega)n) = \cos(\pm\Omega n)$$

- Ω and $-\Omega$ produce the same signals!

Question 2: Sampling Sinusoids

- Our continuous signal is defined as follows:

$$x_c(t) = \cos(2\pi f t)$$

- Now, we sample the signal at $f_s = 2000$ Hz
- What is T_s, and what signal do we obtain?

2(a,b): Sampling the Signal

- The sampling interval is

$$T_s = \frac{1}{2000} \text{ sec} = \boxed{0.5 \text{ ms}}.$$

- Substituting $t = nT_s$ gives

$$x[n] = x_c(n/2000) = \boxed{\cos\left(\frac{\pi f}{1000} n\right)}.$$

2(c): does the observation determine f?

- We observe $x[n] = \cos(\pi f n/1000) = \cos(0.1\pi n)$
- Which other (discrete) frequencies give the same output?

$$\Omega = 2\pi m \pm 0.1\pi$$

- That means:

$$\frac{\pi f}{1000} = 2\pi m \pm 0.1\pi$$

- This gives us the set of possibilites $f = 2000m \pm 100$

2(d): What happens when $0 < f < 1000$ Hz?

- Fortunately, only one candidate lies in this range: $\boxed{f = 100 \text{ Hz}}$.
- Therefore, the continuous signal is

$$x_c(t) = \cos(2\pi f t) = \boxed{\cos(200\pi t).}$$

- We can recover the signal from the discrete samples (Nyquist-Shannon)

2(e): What happens when $0 < f < 2000$ Hz

- In this case, we obtain **two** candidates:

$$\boxed{f = 100 \text{ or } 1900 \text{ Hz}}$$

- Let's substitute $f = 1900$ Hz and see what we get:

$$\begin{aligned} x[n] &= \cos\left(\frac{2\pi f}{2000} \cdot n\right) = \cos\left(\frac{1900\pi}{1000} \cdot n\right) = \cos(1.9\pi n) \\ &= \cos(2\pi n - 0.1\pi n) = \cos(-0.1\pi n) = \cos(0.1\pi n) \end{aligned}$$

- The samples coincide with $f = 100$ (Nyquist-Shannon does not hold)

Question 3

Analyzing Discrete-time Systems

Recall: Causality

- Suppose $x[n]$ and $z[n]$ are signals matching to time m:

$$\forall k \leq m : x[k] = z[k]$$

- For a system T, let $T(x) = y$ and $T(z) = w$
- Then, T is causal if:

$$\forall k \leq m : y[k] = w[k]$$

- Informally, the output does not depend on **future** inputs

Recall: BIBO Stability

- **BIBO** stability means every bounded input must give a bounded output.

$$|x[n]| \leq M \quad \Longrightarrow \quad |(Tx)[n]| \leq C \qquad \text{for all } n.$$

- M and C are finite bounds, independent of n.

Recall: Linearity

- We say a system T is linear if:

$$T(ax + bz) = a\,Tx + b\,Tz.$$

- This must hold for **all** inputs x, z and real constants a, b.
- An important consequence of linearity is $T(0) = 0$

Recall: Time Invariance

- Shift the input by r samples; the output must shift by r too.

$$(Tx)[n] = y[n]$$
$$x_r[n] = x[n-r] \quad \Longrightarrow \quad (Tx_r)[n] = y[n-r].$$

- **LTI** systems are both linear and time invariant.

Question 3: The Discrete-time System

- A discrete-time system on real signals is defined by

$$(Tx)[n] = x[n] + x[n-1] + x[n]x[n-1].$$

- We would like to analyze properties of the above system

3(a): Output with Unit Impulse

- Recall: $\delta[n]$ is 1 at $n = 0$, and 0 elsewhere.
- The product $\delta[n]\delta[n-1]$ is always zero.
- Therefore the last term below is 0

$$(T\delta)[n] = \delta[n] + \delta[n-1] + \delta[n]\delta[n-1] = \delta[n] + \delta[n-1].$$

3(b): Causality

- **Yes.** The output at n depends only on $x[n]$ and $x[n-1]$.
- Therefore, signals matching to m will have matching outputs until m
- Formally, for all $n \leq m$

$$\begin{aligned}(Tx)[n] &= x[n] + x[n-1] + x[n]x[n-1] \\ &= z[n] + z[n-1] + z[n]z[n-1] = (Tz)[n]\end{aligned}$$

3(b): BIBO Stability

- Assume $|x[n]| \leq M$ for every n.

$$\begin{aligned} |(Tx)[n]| &= |x[n] + x[n-1] + x[n]x[n-1]| \\ &\leq |x[n]| + |x[n-1]| + |x[n]x[n-1]| \\ &\leq M + M + M^2 = 2M + M^2. \end{aligned}$$

- **Yes.** This bound is finite and the system is BIBO stable.

3(b): Time Invariance

- Shift the input: $x_r[n] = x[n-r]$

$$\begin{aligned}(Tx_r)[n] &= x[n-r] + x[n-r-1] + x[n-r]x[n-r-1] \\ &= x[(n-r)] + x[(n-r)-1] + x[(n-r)]x[(n-r)-1] \\ &= (Tx)[n-r].\end{aligned}$$

- **Yes.** $(Tx_r)[n] = (Tx)[n-r]$

3(c): Linear and Time-invariance (LTI)

- We already know the system is Time-invariant
- Is it also **linear**?
- Try $v[n] = \delta[n] + \delta[n-1]$ and double this input.

$$(Tv)[1] = v[1] + v[0] + v[1]v[0] = 3,$$
$$(T(2v))[1] = 2v[1] + 2v[0] + (2v[1])(2v[0]) = 8 \neq 2(Tv)[1] = 6$$

- **No.** The system is not linear and hence, not LTI

Question 4

Impulse Responses and Convolution

Recall: Convolution

- Recall the impulse response for an LTI system T:

$$h[n] = (T\delta)[n]$$

- This gave rise to the convolutional formula:

$$(Tx)[n] = (x * h)[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m].$$

Recall: Identifying Causality and Stability

- Recall that an LTI system T is causal if its impulse response h satisfies:

$$\boxed{h[n] = 0 \text{ for } n < 0.}$$

- This is both necessary **and** sufficient
- Furthermore, T is BIBO stable iff:

$$\boxed{\sum_{n=-\infty}^{\infty} |h[n]| < \infty.}$$

4(a): The First System

- Recall that T_1 has impulse response:

$$h_1[n] = \delta[n]$$

- **Causal:** The system is causal. $h_1[n] = 0$ for every $n < 0$.
- **Stable:** The impulse response is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h_1[n]| = 1.$$

Hence, it is BIBO-stable.

4(a): The Second System

- Recall that T_2 has impulse response:

$$h_2[n] = u[n]$$

- **Causal:** The system is causal. $h_2[n] = 0$ for every $n < 0$.
- **Stable:** The impulse response is **not** absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h_2[n]| = \sum_{n=0}^{\infty} 1 = \infty.$$

Hence, it is **not** BIBO-stable.

4(b): Evaluating $T_1 x$

- We can directly evaluate the convolutional formula

$$(T_1 x)[n] = \sum_{m=-\infty}^{\infty} x[m]\delta[n-m] = \boxed{x[n]}.$$

- Informally, unit impulse input produces unit impulse output
- Hence, the system simply outputs the input

4(b): Evaluating T_2x?

- Let's evaluate the convolutional formula again:

$$(T_2x)[n] = \sum_{m=-\infty}^{\infty} x[m]u[n-m] = \boxed{\sum_{m=-\infty}^{n} x[m]}.$$

  as $u[n-m] = 1$ when $n \geq m$.
- Hence, T_2 implements an **accumulator**
- It sums up over past values of the input

4(b): Evaluating $T_2 u$?

- We can evaluate the previous formula:

$$(T_2 u)[n] = \sum_{m=-\infty}^{n} u[m] = \boxed{(n+1)u[n]}.$$

  as for $n \geq 0$, there are $(n+1)$ 1s
- Notice that the system is **not** stable

From Step Response to Impulse Response

- Recall that:
$$\delta[n] = u[n] - u[n-1]$$
- We are given:
$$(Tu)[n] = 3\delta[n-2] + 3\delta[n-k] = s[n]$$
- For an LTI system, let $s[n] = (Tu)[n]$ be its step response.

$$\begin{aligned}(T(u[n] - u[n-1]))[n] &= s[n] - s[n-1] \\ &= 3(\delta[n-2] - \delta[n-3]) \\ &\quad + 3(\delta[n-k] - \delta[(n-k)-1])\end{aligned}$$

4(d): Causality of T

- The impulse response is:

$$h[n] = (T(u[n] - u[n-1]))[n] = 3(\delta[n-2] - \delta[n-3]) + 3(\delta[n-k] - \delta[n-(k+1)])$$

- If $k \geq 0$, all four impulses **delayed**.
- If $k < 0$, then $h[k] = 3$: a response before 0:

T is causal if and only if $k \geq 0$.

4(d): Stability of T

- Let's evaluate absolute summability of h

$$\sum_{n=-\infty}^{\infty} |h[n]| = 3 \sum_{n=-\infty}^{\infty} |(\delta[n-2] - \delta[n-3]) + \delta[n-k] - \delta[n-(k+1)]|$$

- For any one term:

$$\sum_{n=-\infty}^{\infty} |\delta[n-\ell]| = 1$$

- Hence, $\sum_{n=-\infty}^{\infty} |h[n]| \leq 12$
- **Yes.** T is BIBO stable for every k